

1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest change.

Ans :

- 1) `git init`
- 2) `git add .`
- 3) `git status`
- 4) `git commit -m "first commit"`
- 5) `git branch -M main`
- 6) `git remote add origin https://github.com/Shubham-2122/xyz.git`
- 7) `git push -u origin main`

new file

- 1) `git add .`
- 2) `git status`
- 3) `git commit -m "file name"`
- 4) `git push -u origin main`

2. What is the difference between git fetch and git pull?

Answer in 2 lines.

- `git fetch` downloads the latest data from the remote repository to your local system without merging it into your active files, keeping your working directory unchanged.
- `git pull` is a combination of `git fetch` and `git merge`, which automatically updates your current local branch with the changes from the remote server.

3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

1. Log in to GitHub and create a new repository.
2. On your local machine, create a file: `echo "Version Control Test" > test.txt`.
3. Initialize and push:

- `git init`
- `git add test.txt`
- `git commit -m "Add small text file"`
- `git remote add origin <repository-url>`
- `git push -u origin main`

Repository Link: https://github.com/KaranLamoria/Software_Eng

4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

- A Fork is a complete personal copy of someone else's repository that exists in your own account, allowing you to make changes without affecting the original project.
- A Pull Request is a notification sent to the original repository owner to review and merge the changes you have made in your fork or branch.

5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

1. **Create and switch to the new branch:**
`git checkout -b feature-test`
2. **Commit changes on the new branch:**
`git add .`
`git commit -m "Commit on feature-test branch"`
3. **Switch back to the main branch:**
`git checkout main`
4. **Merge the feature branch into main:**
`git merge feature-test`